Qualification Pack

Semiconductor IC Design & Verification Professional

QP Code: MSU/ELE/Q0201

Version: 1.0

NSQF Level: 6

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Contents

MSU/ELE/Q0201: Semiconductor IC Design & Verification Professional

Brief Job Description

On completion of this course, candidates will gain practical skills in IC design and verification through a project-based internship. The curriculum imparts practical skills across the design flow, enabling participants to design and verify ICs. By tackling real-world projects, candidates develop problem-solving abilities and industry readiness. The internship culminates in the design of a microchip fabricated on a stable foundry node, with the functional IP subsequently integrated into a silicon-proven IP repository, providing invaluable exposure to the complete IC design lifecycle.

Personal Attributes

To excel in this IC design project work internship, individuals must possess a strong hands-on mentality, coupled with a keen eye for detail and a systematic approach to problem-solving. Adaptability and a willingness to learn are essential for navigating the dynamic IC design landscape. The ability to work independently and collaboratively within a team is crucial for project success. A strong work ethic, coupled with a passion for technology, is vital for thriving in this fast-paced environment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. MSU/ELE/N0201: Semiconductor Chip Development project briefing, planning and allocation
2. MSU/ELE/N0202: Design Development Methodology Discussion
3. MSU/ELE/N0203: Semiconductor Chip Design Development & Verification
4. MSU/ELE/N0204: Design Characterization
5. MSU/ELE/N0205: Test Chip Development & Tape-out
6. MSU/ELE/N0206: Post Silicon Testing
7. DGT/VSQ/N0103: Employability Skills (90 Hours)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Strategic Electronics
Occupation	Semiconductor IC Design and Verification

GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP

Qualification Pack

Country	India
NSQF Level	6
Credits	23
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3118.0302, 2152.0502
Minimum Educational Qualification & Experience	Completed 4 year UG degree with Honours/ Honours with research OR Completed 3 year UG degree (in relevant field*) with 1.5 years of experience OR Pursuing 4th year UG (in case of 4-year UG with honours/ honours with research) (in relevant field*) OR Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	NG-06-EH-02708-2024-V1-MSU
NQR Version	1

MSU/ELE/N0201: Semiconductor Chip Development project briefing, planning and allocation

Description

This OS unit is about Semiconductor Chip Development project briefing, planning and allocation.

Scope

The scope covers the following :

- Project briefing, planning and allocation

Elements and Performance Criteria

project briefing, planning and allocation

To be competent, the user/individual on the job must be able to:

PC1. explaining the overall semiconductor microchip development program
PC2. design Specification discussion
PC3. semiconductor technology node discussion and MPW (Multi Project Wafer) discussion
PC4. project Allocation and team building

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. a clear overview of the entire semiconductor microchip development program
KU2. about the various stages of the microchip development and the significance of each step
KU3. students will learn how to discuss and define design specifications
KU4. the concept of semiconductor technology nodes
KU5. about Multi Project Wafer (MPW) services and their role in prototyping and reducing costs for multiple projects
KU6. importance of effective team building in achieving project goals and fostering collaboration

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Semiconductor Chip Development project briefing, planning and allocation

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
project briefing, planning and allocation	**100**	**-**	**-**	**-**
PC1. explaining the overall semiconductor microchip development program	20	-	-	-
PC2. design Specification discussion	25	-	-	-
PC3. semiconductor technology node discussion and MPW (Multi Project Wafer) discussion	35	-	-	-
PC4. project Allocation and team building	20	-	-	-
NOS Total	**100**	**-**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0201
NOS Name	Semiconductor Chip Development project briefing, planning and allocation
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design and Verification
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

MSU/ELE/N0202: Design Development Methodology Discussion

Description

This OS unit is about discussing design development methodolgy.

Scope

The scope covers the following :

- Design Development Methodology Discussion

Elements and Performance Criteria

Design Development Methodology Discussion

To be competent, the user/individual on the job must be able to:

PC1. project kickstart meeting- objective and milestone discussion

PC2. explaining the complete flow of the project

PC3. project development methodology discussion

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the importance of a project kick start meeting, learning how to set clear objectives and define key milestones

KU2. gain skills in organising and participating in these meetings to ensure everyone is aligned on the project's goals

KU3. the complete flow of a semiconductor design project

KU4. understand how all the stages are interconnected

KU5. comprehensive knowledge of the PDK used to develop the project

KU6. deeply understand all the methodologies used to design the schematic and layout

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Design Development Methodology Discussion	**100**	**-**	**-**	**-**
PC1. project kickstart meeting- objective and milestone discussion	30	-	-	-
PC2. explaining the complete flow of the project	30	-	-	-
PC3. project development methodology discussion	40	-	-	-
NOS Total	**100**	**-**	**-**	**-**

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0202
NOS Name	Design Development Methodology Discussion
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design and Verification
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

MSU/ELE/N0203: Semiconductor Chip Design Development & Verification

Description

This OS unit is about Design and Verification of Semiconductor IC Macros.

Scope

The scope covers the following :

- Design and Verification of Semiconductor IC Macros

Elements and Performance Criteria

Semiconductor Chip Design Development & Verification

To be competent, the user/individual on the job must be able to:

PC1. schematic Design and Simulation

PC2. physical Layout Design

PC3. DRC (Design Rule Check) and its importance

PC4. DFM – Design for Manufacturability

PC5. LEF (Library Exchange Format) Development

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. about the schematic netlist

KU2. check and correct Design Rule Checking (DRC)

KU3. check and correct for Manufacturability (DFM)

KU4. check and correct Antenna Effects

KU5. expertise in using verification tools and techniques

KU6. about the LEF (Liberty exchange format development)

KU7. skills in simulating IC Macro

KU8. skills in translating IC macro circuit into physical layouts

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Semiconductor Chip Design Development & Verification	**-**	**-**	**90**	**10**
PC1. schematic Design and Simulation	-	-	22	2
PC2. physical Layout Design	-	-	22	2
PC3. DRC (Design Rule Check) and its importance	-	-	22	2
PC4. DFM – Design for Manufacturability	-	-	12	2
PC5. LEF (Library Exchange Format) Development	-	-	12	2
NOS Total	**-**	**-**	**90**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0203
NOS Name	Semiconductor Chip Design Development & Verification
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design and Verification
NSQF Level	6
Credits	11
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

MSU/ELE/N0204: Design Characterization

Description

This OS unit is about about Physical Layout Design Tuning.

Scope

The scope covers the following :

- Physical Layout Design Tuning

Elements and Performance Criteria

Physical Layout Design Tuning

To be competent, the user/individual on the job must be able to:

PC1. LVS – Layout Vs Schematic

PC2. PEX – Parasitic Extraction and .lib introduction

PC3. post Layout Vs Pre-Layout Simulation

PC4. design Tuning

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. introduction to .lib files, which contain the timing and power characteristics of cells

KU2. the importance of Layout Vs Schematic (LVS) checks

KU3. identify and measure parasitic elements like resistance and capacitance

KU4. the differences between pre-layout and post-layout simulations

KU5. to learn how to optimise the physical layout to improve performance

KU6. explore techniques for adjusting the layout to meet design specifications

KU7. to check and correct LVS errors

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Physical Layout Design Tuning	**-**	**-**	**70**	**30**
PC1. LVS – Layout Vs Schematic	-	-	-	-
PC2. PEX – Parasitic Extraction and .lib introduction	-	-	-	-
PC3. post Layout Vs Pre-Layout Simulation	-	-	-	-
PC4. design Tuning	-	-	-	-
NOS Total	**-**	**-**	**70**	**30**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0204
NOS Name	Design Characterization
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design and Verification
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

MSU/ELE/N0205: Test Chip Development & Tape-out

Description

This OS unit is about Test chip design & development and Project Sign-off and Tape-out.

Scope

The scope covers the following :

- Test chip design & development
- Project Sign-off and Tape-out

Elements and Performance Criteria

Test chip design & development

To be competent, the user/individual on the job must be able to:

PC1. Logic Design and Simulation

PC2. floor Planning

PC3. place and Route

PC4. Post Layout Simulation

PC5. Test chip Verification

PC6. introduction to EM IR (Electromigration & IR Drop Analysis)

Project Sign-off and Tape-out

To be competent, the user/individual on the job must be able to:

PC7. sign-off Verification

PC8. checklist review and Tape-out

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the importance of floor planning in chip design

KU2. comparing results with pre-layout simulations to identify and correct any performance issues arising from the layout.

KU3. understanding Electromigration & IR drop and their impact on chip reliability and performance

KU4. learn to place standard cells and route the interconnections to meet design constraints

KU5. learn to simulate the physical design with parasitic elements included

KU6. learn the steps involved in verifying a test chip, including functional verification, timing analysis, and power analysis

KU7. learn the importance of sign-off verification

KU8. understanding of the tape-out process

KU9. learn the steps involved in preparing and submitting the final design for fabrication

KU10. learn the importance of maintaining detailed project documentation throughout the design process

KU11. performing a thorough checklist review before tape-out

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Test chip design & development	**-**	**-**	**50**	**20**
PC1. Logic Design and Simulation	-	-	-	-
PC2. floor Planning	-	-	-	-
PC3. place and Route	-	-	-	-
PC4. Post Layout Simulation	-	-	-	-
PC5. Test chip Verification	-	-	-	-
PC6. introduction to EM IR (Electromigration & IR Drop Analysis)	-	-	-	-
Project Sign-off and Tape-out	**-**	**-**	**20**	**10**
PC7. sign-off Verification	-	-	-	-
PC8. checklist review and Tape-out	-	-	-	-
NOS Total	**-**	**-**	**70**	**30**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0205
NOS Name	Test Chip Development & Tape-out
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design and Verification
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

MSU/ELE/N0206: Post Silicon Testing

Description

This OS unit is about post silicon testing.

Scope

The scope covers the following :

- Post Silicon Testing

Elements and Performance Criteria

Post Silicon Testing

To be competent, the user/individual on the job must be able to:

PC1.	test bench design & development
PC2.	post Silicon Result Analysis
PC3.	CAD Vs Silicon Analysis
PC4.	project closure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. learn how to design and develop a test bench to evaluate the performance of fabricated silicon chips

KU2. learn the steps involved in closing out a semiconductor design project after successful post-silicon testing

KU3. gain practical skills in setting up and configuring test environments

KU4. learn how to interpret test data, identify discrepancies or anomalies, and determine the root causes of any issues found during testing

KU5. learn to identify differences between the designed and fabricated chips

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Post Silicon Testing	**-**	**-**	**45**	**55**
PC1. test bench design & development	-	-	-	-
PC2. post Silicon Result Analysis	-	-	-	-
PC3. CAD Vs Silicon Analysis	-	-	-	-
PC4. project closure	-	-	-	-
NOS Total	**-**	**-**	**45**	**55**

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0206
NOS Name	Post Silicon Testing
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design and Verification
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the current job market requirement and future of work

PC2. identify and explore learning and employability relevant portals

PC3. research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC5. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC6. recognize the significance of 21st Century Skills for employment

PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

PC8. adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC9. use basic English for everyday conversation in different contexts, in person and over the telephone

PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC11. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC12. identify career goals based on the skills, interests, knowledge, and personal attributes

PC13. prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings

PC15. use active listening techniques for effective communication

PC16. communicate in writing using appropriate style and format based on formal or informal requirements

PC17. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC18. communicate and behave appropriately with all genders and PwD

PC19. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.

PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook

PC22. identify common components of salary and compute income, expenses, taxes, investments etc

PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC24. operate digital devices and use their features and applications securely and safely

PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.

PC26. display responsible online behaviour while using various social media platforms

PC27. create a personal email account, send and process received messages as per requirement

PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications

PC29. utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC33. identify different types of customers and ways to communicate with them

PC34. identify and respond to customer requests and needs in a professional manner

PC35. use appropriate tools to collect customer feedback

PC36. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC37. create a professional Curriculum vitae (Résumé)

PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC39. apply to identified job openings using offline /online methods as per requirement

PC40. answer questions politely, with clarity and confidence, during recruitment and selection

PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. components of salary and how to compute income and expenditure
KU12. importance of maintaining safety and security in offline and online financial transactions
KU13. different legal rights and laws
KU14. different types of digital devices and the procedure to operate them safely and securely
KU15. how to create and operate an e- mail account
KU16. use applications such as word processors, spreadsheets etc.
KU17. how to identify business opportunities
KU18. types and needs of customers
KU19. how to apply for a job and prepare for an interview
KU20. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence in English and other languages
GS2. communicate effectively using appropriate language in formal and informal settings
GS3. behave politely and appropriately with all to maintain effective work relationship
GS4. how to work in a virtual mode, using various technological platforms
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
Basic English Skills	**3**	**4**	**-**	**-**
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**5**	**-**	**-**
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

● Batches assigned to the assessment agencies for conducting the assessment on SID or email

● Assessment agencies send the assessment confirmation to VTP/TC looping Madhavi Skills University (MSU)

● Assessment agency deploys the ToA-certified Assessor for executing the assessment

● MSU monitors the assessment process & records

Mode of assessment: - Computerised online assessment theory

● Assessor based practical assessment.

2. Testing Environment:

● Check the Assessment location, date and time

● If the batch size is more than 30 (For a classroom based assessment), then there should be 2 Assessors.

● Check that the allotted time for the candidates to complete Theory & Practical Assessment is correct.

● Computer with internet connectivity

3. Assessment Quality Assurance levels / Framework:

● Question bank is created by the Subject Matter Experts (SME) are verified by the other SMEs

● Questions are mapped to the specified assessment criteria

● Assessor must be ToA certified & trainer must be ToT Certified

● Minimum 50% passing marks

4. Types of evidence or evidence-gathering protocol:

● Time-stamped & geotagged reporting of the candidate from assessment location

● Centre photographs with signboards and scheme-specific branding

5. Method of verification or validation:

● Computerised online randomised objective type questionnaire for theory

● Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

● Soft Copy results are emailed automatically to the concerned authorities and the candidates.

● Hard and copies of the documents are stored

Minimum Aggregate Passing % at QP Level : 50

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MSU/ELE/N0201.Semiconductor Chip Development project briefing, planning and allocation	100	0	0	0	100	8
MSU/ELE/N0202.Design Development Methodology Discussion	100	0	0	0	100	8
MSU/ELE/N0203.Semiconductor Chip Design Development & Verification	0	0	90	10	100	40
MSU/ELE/N0204.Design Characterization	0	0	70	30	100	10
MSU/ELE/N0205.Test Chip Development & Tape-out	0	0	70	30	100	12
MSU/ELE/N0206.Post Silicon Testing	0	0	45	55	100	12
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	**220**	**30**	**275**	**125**	**650**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.